UDSC-502

1. Understanding Data -

a) Brief note on DM & ML and its related branches of study

- Statistics (Descriptive and Inferential)
- Population and Sample

b) Data Matrix and Data Types with examples

- Nominal
- Ordinal
- Interval
- Ratio

c) How to Collect Data (Cases and Variables)

d) Point View and Attribute View

e) The notion of distance (L1 – Manhattan, L2 – Euclidean, and $L\infty$ with general rule) and angle (Cos θ) for both views with examples

f) The geometric, algebraic, statistical, and probabilistic views of the data matrix

- Dot Products
- Centered Views (Mean)
- Orthogonal Projections and their relationship with Mean

g) Dimensionality reduction using PCA

2. Unsupervised Learning:-

a) What is Churning in relation to dataset?

b) How to group the points in Unsupervised Learning?

3. Hyperplanes:-

a) How are hyperplanes modeled as decision boundaries?

b) How are weight vectors orthogonal to hyperplanes?

c) Axis Parallel Hyperplanes and Decision Tree

4. Decision Trees:-

a) What type of ML task is performed by Decision Trees?

b) Explain the working procedure of a Decision Tree classifier with an example.

c) Define a Decision Tree Classifier.

d) What are the components of a Decision Tree and what are decision rules?

e) What is the principle/technique behind the functionality of a Decision Tree classifier? Explain the technique mathematically.

f) Define a split point.

g) Define Data Partition.

h) Define purity in the context of a region/space.

i) How to find the split point in case of a categorical attribute?

j) What is the stopping condition for splitting?

5. Support Vector Machine (SVM):-

a) What type of machine learning task is resolved using SVM?

b) What type of classification task is implemented using SVM?

c) What is the decision boundary in SVM and its mathematical/vector representation?

d) How do you visualize a hyperplane in a 3-dimensional space?

e) Why is it said that the weight vector is normal/orthogonal to the hyperplane? Prove.

f) What are the classes defined in an SVM based on the hyperplane?

g) What is a convex hull?

h) Why do we call the classification task using SVM a large margin classifier?

i) What is the distance of a point to the hyperplane, and how to calculate it?

j) What is the distance of the origin to the hyperplane?

k) Why is the distance calculated called a signed distance?

l) How to convert the signed distance into unsigned distance?

m) What is margin and large margin?

n) What is the objective function of a large margin classifier? Explain necessary transformations.

o) What are the unknown parameters in the objective function?

p) How to solve a primal convex minimization problem with quadratic inequalities using optimization techniques?

q) What do you mean by dual, and how do you realize it in SVM?

r) What are the values of unknown parameters obtained?

s) What is the final result of the SVM classifier for a new point?

6. Soft Margin Classifier (SMC):-

a) What do you mean by soft margin / linearly inseparable?

b) How to handle the misclassified points in SMC?

c) What is the objective function of SMC?

d) What is the meaning of regularization and hinge loss?

e) What are the unknown parameters in the objective function?

f) How to solve a primal convex minimization problem with quadratic inequalities using optimization techniques?

g) What do you mean by dual, and how do you realize it in SMC?

h) What are the values of unknown parameters obtained?

i) What is the final result of the SMC classifier for a new point?

7. Artificial Neural Network (ANN):-

a) What is a perceptron?

b) What type of task is performed by PA and which decision boundary is used for the task?

c) How is the task achieved by the PA?

d) What is the geometric interpretation of PA?

e) Write and explain the PA.

f) How to implement linear and logistic regression models using ANN?

8. Hierarchical Clustering: -

a) What is the goal of Hierarchical Clustering?

b) How do you represent a Hierarchical Clustering?

c) What is nested partition? Explain with an example.

d) What are the different types of Hierarchical Clustering?

e) Write down the algorithm for Agglomerative Clustering. In which fashion does it work?

f) What are the different distance measures for inter-clustering? Show with examples.

g) Explain Single Link Agglomerative Clustering with an example.

9. Density-Based Clustering (DBC):-

a) What is the advantage of Density-Based Clustering (DBC)?

b) What is the DBC approach?

c) Define Core, Border, and Noise Points.

d) Explain DBSCAN in brief.

e) Write down the DBSCAN algorithm.

10.LDA

a. What is Linear Discriminant Analysis (LDA) and in what type of machine learning problem is it used?
b. Explain the main goal of LDA in terms of class separation and dimensionality reduction.
c. What are the key assumptions made by LDA for it to work effectively?
d. How does LDA maximize the separation between different classes in a dataset?
e. Describe the concepts of between-class scatter and within-class scatter matrices in LDA.
f. What is the objective function (criteria) that LDA maximizes to find the best linear discriminant?
g. How is the projection vector for LDA computed using eigenvalues and eigenvectors?

11. PROBABILISTIC CLASSIFIER

a. What is Bayes Theorem and how is it used in classification tasks?
b. Define prior probability, likelihood, and posterior probability with respect to Bayes classifiers.
c. Explain the assumption of feature independence in the Naive Bayes classifier.
d. How is the covariance matrix simplified in the Naive Bayes approach and what does it imply about feature correlation?
e. Describe the difference between full Bayes classifier and Naive Bayes classifier in terms of modelling class distributions.
f. What is the role of the multivariate Gaussian distribution in the Bayes classifier?
g. How are the mean vector and covariance matrix estimated for each class from training data?
h. Explain with an example how a new data point is classified using the Bayes classifier.
i. What is the significance of diagonal covariance matrices in terms of calculating probabilities?

j. How do you compute the posterior probability for a test point given class conditional probabilities and priors?

12: Linear and Logistic Regression

a) What is the objective function of Linear Regression and how to solve it?
b) Explain the geometric interpretation of a bivariate regression model.
c) What is logit function?
d) How to convert a linear model into probabilistic model and what should be the objective function of the probabilistic model?
e) How to classify a data set using logistic regression explain with an example?
f) Explain stochastic gradient algorithm in the context of logistic regression.

13. K-Means and Graph and Spectral Clustering

a) Write down the K- Means algorithm, and explain its working with an example.

b) Write down short notes on Graph and Spectral Clustering.

14. KERNEL METHODS

a) What is advantage of using kernel Methods.

b) Name few kernel method and briefly explain about them.

c) What are operations that can be done in feature space using Kernel Matrix and Kernel Function?

15. Metrics

a) List the Metrics for the following:

i) Regression

ii) Classification

iii) Clustering

16.Frequent item pattern Mining

a) Explain Apriori Algorithm along with its limitations.

(*frequent sequence Mining as short note*)

b) Short note on Maximal, minimal and closed patterns

c) Notion of subgraph pattern mining and its applications

d) What is Association Rule? How to measure the interestingness of such rules?